

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No. 14



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TOPICS TO BE
COVERED

01 Algebraic structures

02 Semi Group

03 Monoid

04 Group

GROUP THEORY

(set, operation)



- +
-
- ÷
- x
- gcd
- lcm

(G, *)



- +
-
- ÷
- x

GROUP THEORY



$(G, *)$ is called group.

- 1. $a \in G, b \in G, a * b \in G$ (closed)
- 2. $a * (b * c) = (a * b) * c$ (Associative)
- 3. $a * e = a$ (Identity property)
 $\swarrow$
 identity element
- 4. $a * a^{-1} = e$ (identity) (Inverse)

Identity

$$a * e = a$$

$$a * e = a$$

Inverse:

$$a * a^{-1} = e(\text{identity})$$

$$a * a^{-1} = e(\text{identity})$$

↙ Inverse

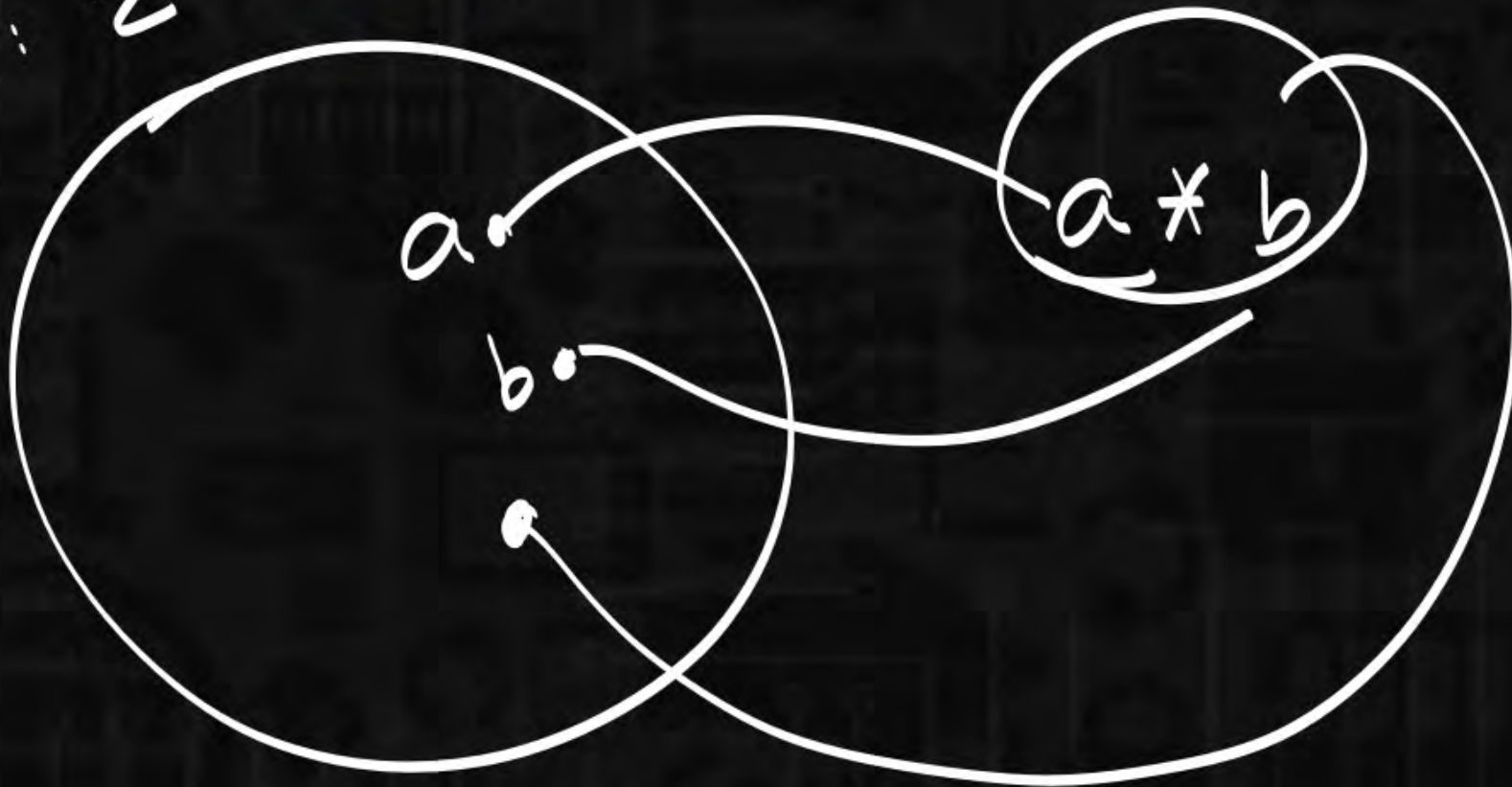
GROUP THEORY



1) closed.

$$a \in G \quad b \in G \quad a * b \in G$$

$G: \mathbb{Z}$
set: $\mathbb{Z}$

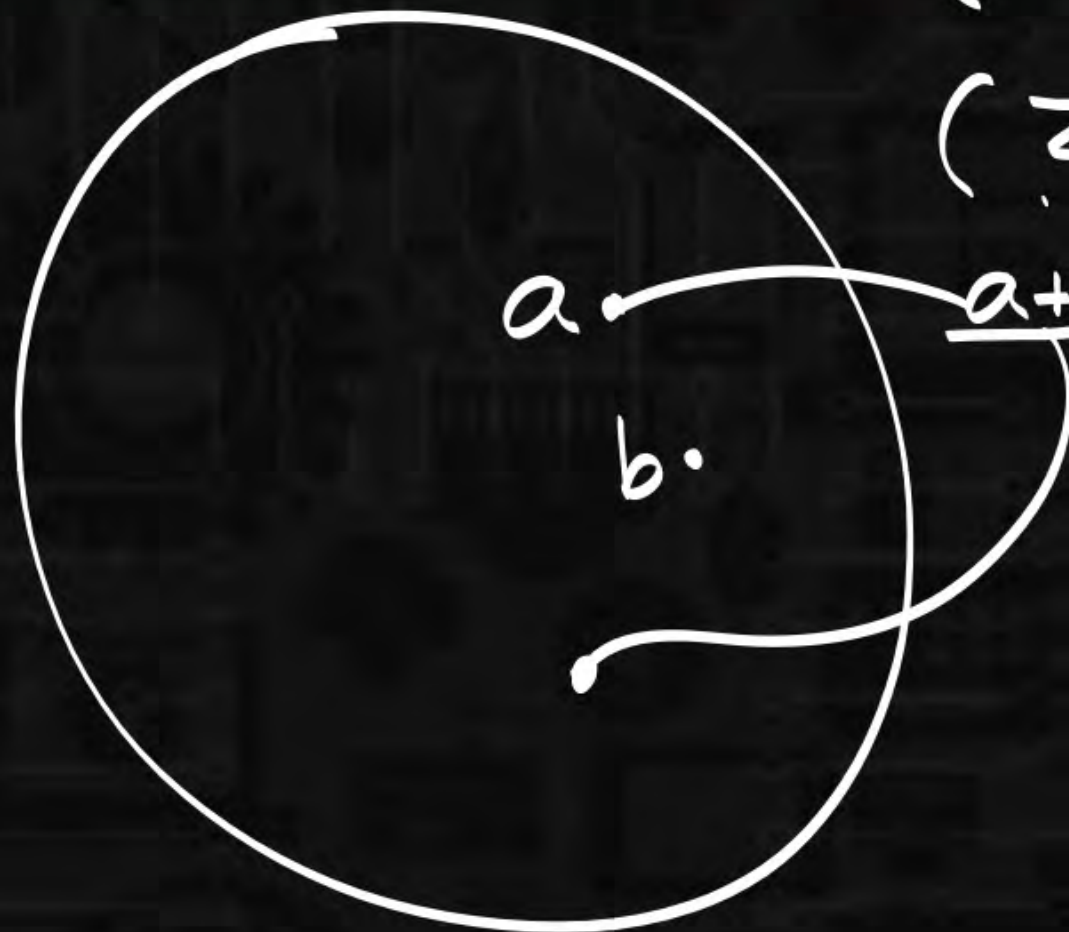


GROUP THEORY



$(G, *)$

$(\mathbb{Z}, +)$



$$a \in \mathbb{Z}$$

$$b \in \mathbb{Z}$$

$$a+b \in \mathbb{Z}$$

$$2 \in \mathbb{Z}$$

$$3 \in \mathbb{Z}$$

$$2+3 \in \mathbb{Z}$$

$$-2 \in \mathbb{Z}$$

$$3 \in \mathbb{Z}$$

$$-2+3 \in \mathbb{Z}$$

GROUP THEORY

$$(G, *)$$

$$\downarrow \quad \downarrow$$

$$(Z, +)$$

$$(Z, +)$$

is called
group.

$$1) a \in Z, b \in Z \quad a+b \in Z \text{ (closed)}$$

$$2) a+(b+c) = (a+b)+c \text{ (Associative)}$$

$$2 + (3+5) = (2+3) + 5$$

$$3) \text{ identity: } \underline{a} + 0 = \underline{a}$$

$$2 + (-2) = 0$$

$$4) \text{ Inverse: } a + (-a) = 0$$

$$3 + (-3) = 0$$

GROUP THEORY

$(G, *)$
 $(\mathbb{Z}, \times)$

not group.

1) $a \in \mathbb{Z} \quad b \in \mathbb{Z} \quad a \times b \in \mathbb{Z}.$ (closed) ✓
 $2 \in \mathbb{Z} \quad 3 \in \mathbb{Z} \quad 2 \times 3 \in \mathbb{Z}.$

2) $a \times (b \times c) = (a \times b) \times c$ (Associative) ✓

3) $a \times 1 = a$ (identity)

4) $a \times \frac{1}{a} = 1$ $\left(\frac{1}{a} \notin \mathbb{Z} \right)$ (Inverse does not exist)

GROUP THEORY



$$(G, *) \quad \underline{a * b} = a + b - ab$$

$1 + 2 - 1 \cdot 2$

$$(\mathbb{Z}, *) \quad 1) \quad a \in \mathbb{Z} \quad b \in \mathbb{Z} \quad \underline{a * b}$$

$$a + b - ab \in \mathbb{Z} \quad (\text{closed}) \checkmark$$

$$2) \quad \underline{(a * b) * c} =$$

$$\underline{(a + b - ab) * c}$$

$$\underline{a + b - ab} + c - c(a + b - ab)$$

$$\underline{a + b - ab + c - ac - cb + abc}$$

$$a + b + c - ab - ac - bc + abc$$

$$a * (b * c)$$

$$\underline{a * (b + c - bc)}$$

$$\underline{a + b + c - bc - a(b + c - bc)}$$

$$\underline{a + b + c - bc - ab - ac + abc}$$

$$a + b + c - ab - bc - ac + abc$$

GROUP THEORY



$$3) \quad a * e = a.$$

$$a + e - ae = a.$$

$$\cancel{a} - \cancel{a} + e - ae = 0$$

$$e = ae.$$

$$a = 1$$

$$3 * e = 3.$$

$$\cancel{3} + e - 3 \cdot \cancel{e} = \cancel{3}.$$

$$e = 3e.$$

$$e = 0$$

GROUP THEORY



$(R, *)$ Check it is group?

$$a^1 * b^2 = \frac{a^1 b^2}{3}$$

1) $a \in R, b \in R, a * b \in R, \frac{ab}{3} \in R.$

2) $a * (b * c) = (a * b) * c$

$(R^+, *)$

$$\bar{a} * \left(\frac{bc}{3} \right) \quad \left(\frac{ab}{3} * c \right)$$

is group.

$$\frac{a * \frac{bc}{3}}{3} = \frac{abc}{9}$$

$$\frac{\frac{ab}{3} * c}{3} = \frac{abc}{9}$$

$(R - \{0\}, *)$

GROUP THEORY



$$a * b = \frac{ab}{3}$$

$$3) \quad a * e = a$$

$$\frac{ae}{3} = b$$

$$\boxed{e = 3}$$

$$4) \quad a * \bar{a}^1 = e$$

$$\frac{a \cdot \bar{a}^1}{3} = e$$

$$a \cdot \bar{a}^1 = 3 \cdot 3$$

$$\bar{a}^1 = \frac{9}{a} (a \neq 0)$$

GROUP THEORY

$$(\mathbb{Z}, *)$$

$$a * b = a + b + 1$$

Group ✓

Closed ✓
 Associative ✓
 $a * e = a$
 $a + e + 1 = a$
 $e = -1$

$$a * a^{-1} = -1$$

$$a + a^{-1} + 1 = -1$$

$$a^{-1} = -1 - 1 - a$$

$$\underline{a^{-1} = -2 - a}$$

Inverse of 3

$$3 * a^{-1} = -1 \quad \underline{3^{-1} = -5}$$

$$3 + a^{-1} + 1 = -1$$

$$a^{-1} = -2 - 3$$

$$\boxed{a^{-1} = -5}$$

$$(\mathbb{Q}_{\neq -1}, *)$$

Group ✓

GROUP THEORY



$$1 + 2 + 1.2$$

1) closed. $a \in \mathbb{Q} \quad b \in \mathbb{Q} \quad a + b + ab \in \mathbb{Q}.$

2) $(a * b) * c = a * (b * c)$

$$(a + b + ab) * \underline{c}$$

$$\rightarrow a * (b + c + bc)$$

$$\underline{a + b + ab} + \underline{c} + ac + bc + abc.$$

$$a + b + c + bc + ab + ac + abc.$$

$$a + b + c + ab + ac + bc + abc$$

GROUP THEORY



3) Identity:

$$\underline{a * e = a}$$

$$\cancel{a} + e + a e = \cancel{a}$$

$$e + a e = 0$$

$$e(1+a) = 0$$

Inver

$$2 * \bar{a}^{-1} = e$$

$$2 + \bar{a}^{-1} + 2 \cdot \bar{a}^{-1} = 0$$

$$\bar{a}^{-1}(1+2) = -2$$

$$\bar{a}^{-1} = \frac{-2}{1+2}$$

$$\bar{a}^{-1} = \frac{-a}{1+a} \quad (a \neq -1)$$

GROUP THEORY



Group.

Infinite Group. :

$$(\mathbb{Z}, +)$$

$$(\mathbb{Z}, *)$$

$$a * b = a + b + 1.$$

finite Group.

(Set, operation)

$$(\{ \dots \}, *)$$

$*$	e_1	e_2	e_3
e_1			
e_2			
e_3			

GROUP THEORY

(Set, *)

$$a * b = (a + b) \bmod 6.$$

$(\{0, 1, 2, 3, 4, 5\}, *)$

$$(1 + 5) \bmod 6.$$

$$6 \bmod 6 = 0.$$

$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

GROUP THEORY



(Set, +)

($\{-1, 0, 1\}$, +)

It is not group.

+	-1	0	1
-1	-2	-1	0
0	-1	0	1
1	0	1	2

1) closed. X

$$-1 \in G$$

$$-1 \in G$$

$$-1 + -1 \notin G$$

GROUP THEORY



Set: $\{0, 1, 2, 3, 4, 5\}$

1) Closed ✓

2) Associative ✓

$$2 \oplus_6 (4 \oplus_6 3) = (2 \oplus_6 4) \oplus_6 3$$

$$2 \oplus_6 (1)$$

$$= 3$$

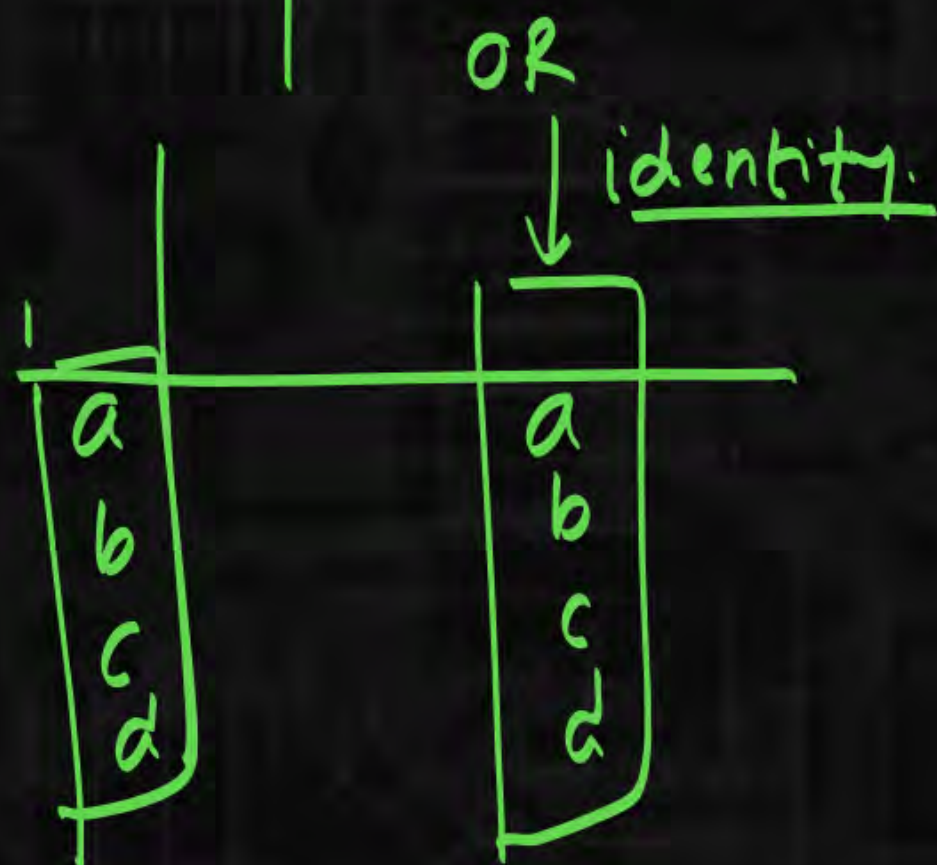
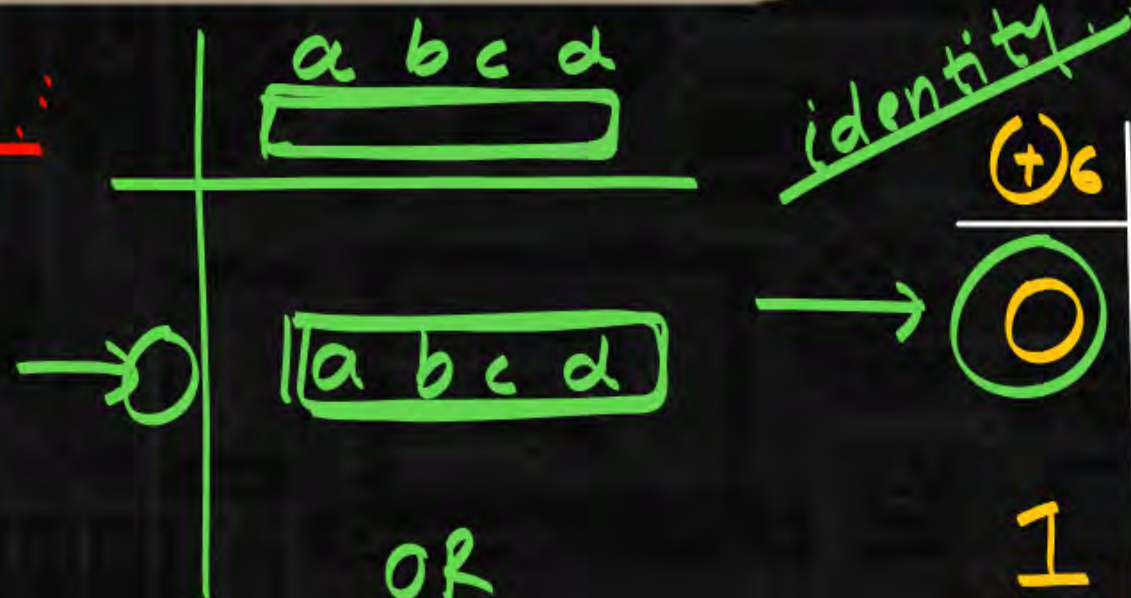
$$0 \oplus_6 3$$

$$= 3$$

$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

GROUP THEORY

3) identity:



$(+)_6$

0

1

2

3

4

5

0	1	2	3	4	5
0	1	2	3	4	5
1	2	3	4	5	0
2	3	4	5	0	1
3	4	5	0	1	2
4	5	0	1	2	3
5	0	1	2	3	4

GROUP THEORY



identity element $\rightarrow 0$

Inverse:

$$\left\{ \begin{array}{l} 2' = 4 \\ 3' = 3 \\ 4' = 2 \\ 5' = 1 \\ 0' = 0 \end{array} \right.$$

Group

mod 6.

$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

GROUP THEORY

(Set, *)

$(\{1, -1, i, -i\}, \times)$

$$i^0 = 1$$

$$i^2 = -1$$

$$i^3 = i^2 \times i$$

$$= -1 \times i = -i$$

$$i^4 = i^2 \times i^2$$

$$= -1 \times -1 = 1$$

$$i^5 = i^4 \times i = 1 \times i = i$$

$$\begin{aligned} i \times -i &= -i^2 \\ &= -(-1) \\ &= 1 \end{aligned}$$

$$i^{-1} = -i$$

$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

